

Weigh and Log Parts SOP

One episode weighs ten parts. It runs from ten parts standing in a numbered rack, an empty scale reading zero, three empty bins and a blank log sheet, to all ten parts weighed one at a time, ten weights and ten band marks written on the sheet, seven parts in the GOOD bin, one part in the TOO LIGHT bin, two parts in the TOO HEAVY bin, the scale empty and reading zero again, and every tool back on its spot.

One part is finished before the next one is started. The order for each part never changes: **take the part out of its slot, set it on the scale, read the weight, write it on the sheet, drop the part in its bin, zero the scale.** No part goes on the scale until the scale reads zero with nothing on it, and no part goes in a bin until its weight is on the sheet.

One arm does all of it. The same gripper takes every part out of the rack, sets it on the scale, lifts it off again, works the pen, presses the ZERO bar and drops the part in its bin. Nothing in this task has to be held, because nothing moves on its own: the rack, the scale, all three bins and the log sheet grip the table, and a part left on the scale pan stands there on its own. Only one arm works in this cell. Do not bring a second arm into the task.

Every spot sits inside the arm's comfortable reach, so the arm never has to stretch or fold in on itself. Only one part is held at a time, no grasp is changed while a part is off the table, and no part is ever carried over a bin it is not going into. Every part leaves the scale to the right, out to the one bin its band says.

The scale does the deciding, not the eye. A part is heavy or light only because the number on the display says so, and the band strip on the sheet says which band that number falls in. Nothing is weighed by hand, no two parts are compared against each other, and no part is judged by how it looks.

The sheet is written in the plainest words the job allows. A part is not logged as out of specification or off nominal: it is marked **TOO LIGHT**, **GOOD** or **TOO HEAVY**, and the number the scale showed is written beside it. Anyone who can read those three words can read the sheet.

Each tool goes back to its own spot before the next step starts. The counts are fixed: ten parts, ten weighings, ten weights written, ten band marks, eleven zeroings, seven parts in the GOOD bin, one part in the TOO LIGHT bin and two parts in the TOO HEAVY bin. There is no second weighing of the same part and no second reading of the same number. A part that lands off the table is left exactly where it falls, tagged, and the episode continues. Nothing is ever picked up off the floor on camera.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** single_arm

Hardware checklist

1. Cameras are on and recording.
2. The environment camera frame holds the whole tabletop in one shot: the parts rack, the scale with its pan and its display, all three bins, the log sheet and the pen holder.
3. The scale display stays in frame and readable the whole time, so a reviewer can read the same number the sheet was written from.
4. The scale pan stays in frame the whole time, so a reviewer can see whether the pan is empty or has a part on it.
5. All ten rows of the log sheet and the band strip at its head stay in frame and readable the whole time.
6. The arm is at home with its gripper open.
7. The tabletop is dry and clear of anything but the items in the Materials checklist.
8. The arm reaches every spot on the table, all ten rack slots, the scale pan, the ZERO bar, all three bins, the log sheet and the pen holder, without extending to a joint limit or folding in on itself. Bring the spots forward or closer together until this is true.
9. The arm reaches every one of the ten rack slots straight on, so a part can be lifted straight up out of any slot without leaning it over.
10. The arm comes down onto the scale pan straight from above, so a part can be set down without being tilted, swung in or slid across the pan.
11. The paths from each rack slot to the scale pan, from the scale pan out to each of the three bins, and from the pen holder to the sheet are unobstructed. Carry one part along each path by hand during setup: it must clear the rim of the scale pan and the rim of every bin it passes.
12. Nothing leans on the scale, touches the scale or rests against it. Even a cable lying across the scale changes what it reads.
13. The table does not rock. Push down on the tabletop beside the scale during setup: if the display moves off zero, steady the table or move the scale.

Materials checklist

1. One parts rack stands at the back of the table, straddling the midline, with ten upright slots in two rows of five, numbered 1 to 5 across the back row and 6 to 10 across the front row, left to right. The numbers face the camera.
2. The rack grips the table. Push it by hand during setup: if it slides, turns or lifts, weight it or replace it. Nothing pins the rack during the episode.

3. Each slot holds one part standing upright with enough clearance that the part lifts straight out without rubbing.
4. Ten parts, and no more. Each part is a plain solid block with flat faces, no burrs and no markings on it. Every part is the same on the outside, so the outside tells a reviewer nothing and only the scale does.
5. Each part is small enough to sit whole inside the scale pan with a clear gap all the way round, and tall enough for the gripper to close on its sides squarely.
6. The ten parts are the same ten in every episode: seven good parts, one light part and two heavy parts.
7. The three bad parts are bad by a plain amount, at least 5 grams outside the good band, so no reading ever lands on the edge of a band. A part that only just fails is not used.
8. One bench scale sits at the middle of the table, straddling the midline, with its pan level and its display facing the camera. It is heavy and it grips the table.
9. The scale reads in whole grams and it reads to at least 100 grams. The display is big enough to read on camera from the environment camera.
10. The scale has one flat ZERO bar on its front face, wide enough to press with a closed gripper, labelled ZERO facing the camera. It is the only control that is ever touched.
11. The scale pan is empty, clean and dry at the start of the episode. A crumb or a chip left in the pan is weight the next part gets blamed for.
12. The scale is on and warmed up, and with nothing on the pan it reads zero and holds there for 5 seconds without drifting.
13. Three bins stand in their holders down the right side of the table, in a line back to front: the TOO LIGHT bin at the back, the GOOD bin in the middle and the TOO HEAVY bin at the front. They are empty and dry, they are clearly different in colour, each is labelled TOO LIGHT, GOOD or TOO HEAVY facing the camera, and the same bin is the GOOD bin in every episode.
14. A clear gap of at least a gripper's width sits between each pair of bins, so a part comes down into one bin without passing over another.
15. Each bin is low enough and wide enough that a part can be dropped in from just above the rim, and each one sits in its holder so it cannot tip or slide when a part lands in it.
16. No bin touches the scale and no bin shares the scale's own surface. A bin knocked while a part is on the pan changes the reading.
17. One log sheet lies flat at the front centre on the midline. It is a wipe clean sheet and it starts the episode blank.
18. The sheet carries ten rows stacked down it, numbered 1 to 10 down the left, matching the ten rack slots. Each row holds one open weight box on the left and three band boxes on the right.

19. The three band boxes in every row are labelled across the top of the sheet in plain words: TOO LIGHT, GOOD, TOO HEAVY. Every box is big enough to hold one X with room to spare, and the boxes do not touch each other.
20. The sheet carries one band strip at its head, printed in full plain words: TOO LIGHT is under 48 grams, GOOD is 48 to 52 grams, TOO HEAVY is over 52 grams. These are the only numbers printed on the sheet.
21. The words on the sheet are printed in full and in plain English. No code, no part number and no specification language appears anywhere on it.
22. The sheet grips the table. Push it by hand during setup: if it slides, tape it down or put it in a holder.
23. One dry erase pen stands in its holder on the sheet's left edge, cap off, with enough ink to write ten weights and ten marks.

Workspace layout

Three rows and one right hand column: the parts at the back, the scale in the middle, the log sheet at the front, and the three bins in a line down the right side. Every part travels forward out of the rack onto the scale, then out to the right into one bin.

Back row:

1. Parts rack, back centre on the midline, ten upright slots in two rows of five, slots 1 to 5 across the back row and slots 6 to 10 across the front row, left to right, with the numbers facing the camera.

Middle row:

2. Scale, centre, straddling the midline, its pan level and empty at the start of the episode and empty again at the end, its display facing the camera and its ZERO bar on the front face.

Front row, left to right:

3. Pen holder, on the log sheet's left edge.
4. Log sheet, front centre on the midline, its ten rows and its band strip facing the camera.

Right column, back to front:

5. TOO LIGHT bin, back right, standing in its holder.
6. GOOD bin, middle right, standing in its holder, straight out to the right of the scale.
7. TOO HEAVY bin, front right, standing in its holder.

A clear gap of at least a gripper's width sits between the scale and the nearest bin, between each pair of bins, and between the scale and the log sheet, so a part is set on the pan without the gripper touching a bin, and drops into a bin straight from above without the gripper touching the scale. Carry a part forward or back on the scale's own line first, then straight out to the right into the mouth of its bin. No carry passes over a bin the part is not going into.

Vocabulary

- **Arm assignment:** one gripper does everything: every part, every press of the ZERO bar, every mark. This is a single arm task and no second arm takes any part in it.
- **Tool home:** the spot a tool returns to and sits on before the next step starts: the pen holder. A tool is never left on the sheet, on the scale, in a bin or in the gripper between steps.
- **Release point:** the gripper opens over the middle of the scale pan when a part is resting on it, just above the rim of a bin when a part is being dropped in, or over a tool home. Never over the wrong bin and never over open tabletop.
- **Stable placement:** the object stands still after release, square on its surface, and does not rock, tip, slide or lean on anything.
- **Part:** one plain solid block. There are ten in every episode, one to a rack slot.
- **Scale:** the bench scale at the middle of the table. It is the only thing in this task that decides anything.
- **Pan:** the flat tray on top of the scale that a part is set on. Only one part sits on it at a time, and nothing else is ever set on it.
- **Display:** the window on the scale that shows the weight in whole grams. It faces the camera and it is never blocked while a number is being read.
- **ZERO bar:** the flat bar on the front of the scale. Pressing it once sets the display back to zero. It is the only control that is ever touched.
- **Zeroed:** the pan is empty and the display reads zero. The scale is zeroed before every part.
- **Steady reading:** the number on the display has stopped changing and has held the same for 2 seconds with nothing touching the part, the pan, the scale or the table.
- **Band strip:** the line at the head of the sheet that says which weights are TOO LIGHT, which are GOOD and which are TOO HEAVY. It is read, never written on.
- **Band:** the one word the band strip gives a steady reading: TOO LIGHT, GOOD or TOO HEAVY. Every part has exactly one band.
- **TOO LIGHT:** the steady reading is under 48 grams. The part weighs less than it should. It goes in the TOO LIGHT bin.
- **GOOD:** the steady reading is 48 to 52 grams. The part weighs what it should. It goes in the GOOD bin.
- **TOO HEAVY:** the steady reading is over 52 grams. The part weighs more than it should. It goes in the TOO HEAVY bin.
- **Row:** one of the ten lines on the sheet. A part's row is the row with its own slot number on it.
- **Weight box:** the open box at the left of a row. It holds the number the display showed, written in full, and nothing else.
- **Band box:** one of the three boxes at the right of a row. Exactly one of them carries an X, and it is the one the band strip says.

- **Marked box:** a box on the sheet holding its mark, fully inside its lines and not touching the next box.
- **One part at a time:** the gripper takes one part out of the rack, weighs it, writes it down and bins it before it touches the next part.
- **Empty pan:** nothing on the pan and nothing leaning on the scale.
- **Spill:** a part outside the rack, the pan and the three bins. It is left where it falls and cleared in the reset, never on camera.
- **Quarter turn:** the most any object is tilted in the gripper. Every part travels level, and nothing is ever turned over in the air.

Steps

Step 1: Zero the scale

Goal: the pan empty, the ZERO bar pressed once, and the display reading zero.

The scale is zeroed before the first part and again after every part. A scale that is not at zero puts the last part's weight into this part's number, so nothing goes on the pan until the display reads zero with nothing on it.

1.1 Check the pan

- **Check:** the pan is empty, clean and dry, with nothing leaning on the scale.
- Nothing is brushed off the pan with the gripper. A pan with something in it is a reset job.

1.2 Press the ZERO bar

- With the **gripper** closed, come down onto the ZERO bar straight from the front, press it once, and come away.
- One press. Do not hold the bar down and do not press it twice.
- Do not lean on the scale body and do not touch the pan.

1.3 Check the display

- Hold still and wait 2 seconds with nothing touching the scale or the table.

Check: the display reads zero and holds there. If it does not read zero, or it drifts off zero, end the episode and correct the scale during reset. Do not press the bar again to chase it.

Step 2: Take the part from its slot

Goal: one part out of its slot, hanging level in the gripper, and its slot empty.

Parts are worked in slot order, slot 1 first and slot 10 last. The rack and the scale both grip the table, so nothing holds either of them while the part is moved.

2.1 Lift the part straight out

- With the **gripper**, close on the sides of the part square, and lift it straight up out of the slot until it is clear of the rack.
- Keep it level. Do not lean it over on the way out.

Check: the part hangs level in the gripper and does not slip. If it hangs tilted, stand it back in its slot and take it again. Make no more than two corrective takes. If it still will not hang level, end the episode and correct the part during reset.

Step 3: Weigh the part

Goal: the part standing on its own in the middle of the pan, let go of, and one steady number on the display.

The part is set down and then released. It is never held while the number is being read, because a hand still on the part is weight the scale is reading too.

3.1 Set the part on the pan

- Carry the part forward to the scale, keeping it level and clear of the display, and bring it over the middle of the pan.
- Lower it straight down slowly until it takes its own weight on the pan, then stop.
- Do not drop it onto the pan, do not slide it across the pan, and do not press it down.
- The part sits whole inside the pan, with a clear gap all the way round it.

Check: the part stands square and level in the middle of the pan, touching nothing but the pan, and no part of it hangs over the rim.

3.2 Let go and wait for the number to stop

- Open the gripper and take it right away from the part, the pan and the scale.
- Hold still and wait 2 seconds with nothing touching the part, the scale or the table.
- The number on the display has to stop changing and hold the same for those 2 seconds.
- Do not steady the part, do not tap the pan and do not press the ZERO bar with a part on it.

Check: the display is readable in frame, the arm is clear of it, and the number has held still. If the number is still moving after 5 seconds, end the episode and let review flag it.

3.3 Read the band off the strip

- Read the steady number on the display.
- **If the number is under 48 grams**, the part is **TOO LIGHT**.
- **If the number is 48 to 52 grams**, the part is **GOOD**.
- **If the number is over 52 grams**, the part is **TOO HEAVY**.
- Leave the part standing on the pan. It stays there while the sheet is written.

The band strip at the head of the sheet says the same three lines. It is read, never written on, and it is never worked out in the head.

Check: the part is still on the pan and the number on the display has not changed.

Step 4: Write it on the sheet

Goal: the number in the weight box of the part's own row, one X in the band box the strip says, and the pen back in its holder.

The part is still standing on the pan while the sheet is written, so the number being copied is still on the display to be seen.

4.1 Write the weight

- With the **gripper**, close on the pen barrel, lift it straight up out of its holder, and bring the tip over the sheet.
- Find the row with the same number as the slot the part came out of.
- Write the number the display is showing in that row's weight box, in full, fully inside the box.
- Copy the number exactly. Do not round it, shorten it or work anything out from it.

One number per row, written once, in the row with the part's own slot number on it.

Check: exactly one new number is on the sheet, it is in the right row, it matches the display, and it is inside its box.

4.2 Mark the band box

- In the same row, draw one X in the band box the strip says: **TOO LIGHT**, **GOOD** or **TOO HEAVY**.
- One X, in one box, in the same row as the number just written.
- Draw it fully inside the box lines, not touching the box beside it.

The number says what the part weighed and the X says which bin it is going in. Nothing else is written on the sheet.

Check: the row carries one number and one X, the X is in the band the number falls in, and the other two band boxes in that row are blank.

4.3 Return the pen to its holder

- With the **gripper**, carry the pen back to its holder, lower it in until it stands on its own, release, and lift clear.

Check: the pen stands in its holder and the sheet carries one number and one X for every part weighed so far. A wrong mark is not rubbed out and not written over. Leave it and let review flag the episode.

Step 5: Put the part in its bin

Goal: the part in the bin its band box says, and the pan empty again.

All three bins stand in their holders, so a part can be dropped into any one of them without anything holding it. No bin is ever lifted.

5.1 Lift the part off the pan

- With the **gripper**, close on the sides of the part square, and lift it straight up off the pan until it is clear of the pan rim.
- Do not drag it across the pan and do not knock the scale on the way up.

Check: the pan is empty and the gripper has not knocked the scale.

5.2 Drop it in its bin

- If the row is marked **TOO LIGHT**, carry the part out to the **TOO LIGHT bin** at the back right.
- If the row is marked **GOOD**, carry the part out to the **GOOD bin** at the middle right.
- If the row is marked **TOO HEAVY**, carry the part out to the **TOO HEAVY bin** at the front right.
- Move the part forward or back on the scale's own line first, then straight out to the right into the mouth of its bin.
- Bring it over the bin mouth, low to the rim, stop, and open the gripper so it drops straight in.
- A part never passes over a bin it is not going into.

Check: the pan is empty and the part is in the bin its row says. If a part has gone into the wrong bin, leave it there and let review flag the episode. Parts are never taken back out of a bin.

Step 6: Zero the scale again

Goal: the pan empty and the display reading zero, ready for the next part.

- Run Step 1 again: check the pan is empty, press the ZERO bar once, and wait 2 seconds.
- The scale is zeroed after every part, including the last one, so the episode ends with the display at zero.
- The ZERO bar is never pressed with a part on the pan.

Check: the pan is empty and the display reads zero and holds. If it will not sit at zero, end the episode and correct the scale during reset.

Step 7: Do the next part

- Come back empty to the rack and take the part from the next slot in order.
- Run Steps 2 to 6 again for that part, writing it in the row with its own slot number on it.

- Ten parts, worked one at a time from slot 1 to slot 10. A part is finished and in a bin before the next part is touched.
- One part on the pan at a time. A second part never goes on with the first one still there.

Check: after each part, the rack has one fewer part in it, the pan is empty, the display reads zero, one more row on the sheet is filled in, and no part is weighed twice or skipped.

Step 8: End the episode

- Confirm the rack: all ten slots empty.
- Confirm the scale: the pan empty and the display reading zero.
- Confirm the bins: seven parts in the GOOD bin, one part in the TOO LIGHT bin, two parts in the TOO HEAVY bin, all three bins upright in their holders.
- Confirm the sheet: all ten rows filled in, each with one number in its weight box and one X in one band box, every mark inside its own box.
- Confirm the marks match: seven rows marked GOOD, one row marked TOO LIGHT and two rows marked TOO HEAVY, and every X in the band its own number falls in.
- Confirm the tools: the pen upright in its holder.
- Correct a failed check only with the correction written in its step. If no correction is written, or the correction would need an earlier step run again, end the episode and reset instead.
- Return the arm home with its gripper open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the parts out of all three bins and stand the bins back in their holders, empty and dry.
2. Count the parts back into the rack: exactly ten, seven good, one light and two heavy. Replace any part that has been chipped, dented or burred so the ten are the same ten next time.
3. Shuffle the ten parts into the ten slots so they do not sit in the same order twice in a row, then stand each one upright.
4. Pick up any part that fell off the pan or off the table and put it back in the rack. Replace it if it is damaged.
5. Wipe the log sheet clean and dry it, so all ten rows and all thirty band boxes start blank.
6. Wipe the scale pan clean and dry. A crumb or a chip left in the pan is weight the next part gets blamed for.
7. Check the scale by hand: with nothing on the pan it must read zero and hold for 5 seconds without drifting.
8. Check the scale against a known weight by hand and confirm it reads that weight. Replace or service a scale that does not.

9. Confirm nothing leans on the scale, touches it or rests against it, and that the table does not rock.
10. Stand the pen in its holder with the cap off. Replace a pen that is running dry.
11. Confirm the rack, the scale, all three bins and the log sheet all still grip the table.
12. Confirm the sheet is still printed in full plain words, with TOO LIGHT, GOOD and TOO HEAVY readable, the band strip readable at its head, and the ten rows still numbered 1 to 10.
13. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the start timestamp, violation name, and SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** a part goes on the pan before the scale reads zero, a part goes in a bin before its weight is on the sheet, the band box is marked before the number is written, a new part is taken while the last one is still on the pan, or a step is skipped.
- **SOP rule broken:** Steps 1 to 7, zero the scale, then for each part take it, weigh it, write it, bin it and zero again, in that order.
- **Coaching note:** zero first, then one part all the way through. The number goes on the sheet while the part is still on the pan.

Violation: A second arm used

- **Visible cue:** a second arm enters the work area, closes its gripper, touches an object, steadies the rack, the scale, a bin or the sheet, or moves in front of the display or the sheet. Any object taken or carried by a second gripper is a violation on its own.
- **SOP rule broken:** Steps 1 to 8, this is a single arm task: one gripper does all of it and nothing else in the cell touches the task.

- **Coaching note:** one arm only. If a second arm has worked, the episode is a bimanual episode wearing the wrong label.

Violation: Object held or steadied

- **Visible cue:** the gripper presses on the rack, the scale, a bin or the sheet to keep it still while working, or holds the part on the pan while the number is being read, instead of letting the pan do it.
- **SOP rule broken:** Steps 1 to 5, every object grips the table or stands in its holder, and nothing on this station is held.
- **Coaching note:** if something needs holding, the fixture is wrong. Fix the station, do not add a hand.

Violation: Scale not zeroed, or zeroed with a part on it

- **Visible cue:** a part goes on the pan with the display showing anything but zero, the ZERO bar is pressed while a part is on the pan, the bar is pressed twice or held down, or the zero is not checked on camera before the part is set down.
- **SOP rule broken:** Steps 1 and 6, the pan is empty, the ZERO bar is pressed once, and the display is seen reading zero before every part.
- **Coaching note:** zero on an empty pan, once, every time. A zero taken with a part on the pan hides the part you are trying to weigh.

Violation: Reading taken before it settles

- **Visible cue:** the number is read or written while the display is still changing, before the gripper has come away, or before the number has held the same for 2 seconds.
- **SOP rule broken:** Step 3.2, the reading is the number the display holds once the part has been let go and left still for 2 seconds.
- **Coaching note:** wait the two seconds. A number still moving has not finished being a weight yet.

Violation: Gripper still on the part while it is weighed

- **Visible cue:** the gripper stays closed on the part, rests on it, on the pan, on the scale body or on the table beside the scale, or nudges any of them while the number is being read.
- **SOP rule broken:** Step 3.2, open the gripper and take it right away before the number is read.
- **Coaching note:** let go, back off, then read. A hand on the part is weight on the scale.

Violation: Scale touched, leaned on, or knocked

- **Visible cue:** the gripper, a carried part or a cable presses on the scale body, drags across the pan, knocks the pan, or the table is jarred while a number is being read.

- **SOP rule broken:** Hardware checklist and Steps 3 and 5, nothing touches the scale while it is being read, and the pan carries the part and nothing else.
- **Coaching note:** the scale reads everything it feels. Keep the whole path clear of it.

Violation: Part set down badly on the pan

- **Visible cue:** the part is dropped onto the pan from height, slid across it, pressed down onto it, set down crooked so it rocks, set down off centre, or set down hanging over the pan rim or touching something outside the pan.
- **SOP rule broken:** Step 3.1, lower the part straight down slowly into the middle of the pan until it takes its own weight, square and level, with a clear gap all the way round.
- **Coaching note:** square it over the pan, then lower it. A part touching anything but the pan is not being weighed on its own.

Violation: More than one part on the pan

- **Visible cue:** a second part is set on the pan before the first is lifted off, the gripper closes on two parts at once, or a tool, the pen or anything else is set on the pan.
- **SOP rule broken:** Steps 3 and 7, one part on the pan at a time, and nothing but a part ever goes on it.
- **Coaching note:** one part, one number. Two things on the pan give one weight that belongs to neither.

Violation: Part weighed twice, or a weighing skipped

- **Visible cue:** a part goes back on the pan after its row is written, a part is lifted and set down again to get a different number, or a part is binned with no weighing on camera.
- **SOP rule broken:** Steps 3 and 7, one weighing per part, once, on every part.
- **Coaching note:** one trip across the pan. A second weighing is the model shopping for a better number.

Violation: Number copied wrong, rounded, or worked out

- **Visible cue:** the number written in the weight box is not the number on the display, it is rounded, shortened, guessed, or a weight is arrived at by adding or taking away instead of read.
- **SOP rule broken:** Step 4.1, copy the number the display is showing, in full, exactly as it reads.
- **Coaching note:** copy, do not calculate. A number that is not on the display did not come from the scale.

Violation: Wrong band box marked

- **Visible cue:** the X goes in a band the number does not fall in, two band boxes are marked for one part, no band box is marked, or the band is chosen by how the part looks rather than off the band strip.
- **SOP rule broken:** Step 4.2, one X per row, in the TOO LIGHT box, the GOOD box or the TOO HEAVY box, chosen by the number against the band strip.
- **Coaching note:** under 48 is TOO LIGHT, 48 to 52 is GOOD, over 52 is TOO HEAVY. Nothing else.

Violation: Wrong row written

- **Visible cue:** the number or the X goes in a row that is not the row with the part's slot number on it, or two rows are written for one part.
- **SOP rule broken:** Steps 4.1, 4.2 and 7, a part is written in the row with its own slot number, and parts are worked slot 1 through slot 10 in order.
- **Coaching note:** the row number is the slot number. A number in the wrong row loses which part it belongs to.

Violation: Parts worked out of slot order

- **Visible cue:** a part is taken from a slot out of turn, or a slot is skipped and come back to later.
- **SOP rule broken:** Steps 2.1 and 7, parts are worked slot 1 through slot 10 in order.
- **Coaching note:** left to right, back row then front row. In order, every time.

Violation: Mark rubbed out, written over, or outside its box

- **Visible cue:** a number or an X is wiped off, written over, corrected, or a mark crosses or sits outside a box line so it cannot be read.
- **SOP rule broken:** Steps 4.1 and 4.2, each mark is one number or one X fully inside its own box, and a wrong mark is tagged, never rubbed out.
- **Coaching note:** leave the mistake and let review flag it. A corrected sheet cannot be trusted.

Violation: Part put in the wrong bin

- **Visible cue:** a part goes in a bin its row does not say, or a part is left on the pan, on the sheet or on the table instead of a bin.
- **SOP rule broken:** Step 5.2, the part goes in the bin its own row on the sheet says.
- **Coaching note:** read the row, then move. The sheet decides the bin, not the eye.

Violation: Part taken back out of a bin

- **Visible cue:** a part is lifted back out of a bin, or a bin is tipped, to move a part that was put in the wrong place.
- **SOP rule broken:** Step 5.2, parts are never taken back out of a bin, and a wrong bin is tagged, not fixed.
- **Coaching note:** once it is in, it stays in. A correction pick makes the episode unlearnable.

Violation: Part released before it is over its bin

- **Visible cue:** the gripper opens while the part is still over the scale, over open tabletop, or over the wrong bin, or it opens high so the part bounces out.
- **SOP rule broken:** Step 5.2, the gripper opens low over the mouth of the bin the part belongs in.
- **Coaching note:** carry it all the way, low to the rim, then open. A part dropped early is a part on the floor.

Violation: A part carried over the wrong bin

- **Visible cue:** a part passes over the top of a bin it is not going into on its way to its own bin, or is carried diagonally across the line of bins.
- **SOP rule broken:** Step 5.2 and Workspace layout, move on the scale's own line first, then straight out to the right into the bin's mouth.
- **Coaching note:** route around, not across. A part dropped over the wrong bin lands in the wrong bin.

Violation: Part rolled, dragged or turned over

- **Visible cue:** the part is rolled or dragged across the pan or the table, tipped onto its side, or rolled past a quarter turn or turned over in the air.
- **SOP rule broken:** Steps 2 to 5, every part travels level, lifted clear of the surface before any horizontal motion.
- **Coaching note:** lift it, carry it level, set it down. A dragged part scratches the pan and adds grit to the next weight.

Violation: Tool not returned to its home

- **Visible cue:** the pen is left lying on the sheet, on the scale, in a bin or on the table, the gripper opens while the pen is still above its holder, or a tool is still in the gripper when the next step begins.
- **SOP rule broken:** Step 4.3, the pen returns to its holder before the next step starts.
- **Coaching note:** the return is part of the step, not a gap between steps. Lower until it touches, then release.

Violation: Spill cleaned up

- **Visible cue:** after a part lands on the pan edge, on the table or on the floor, the gripper is used to sweep, push, or pick it up.
- **SOP rule broken:** Steps 1 to 7, a spill is left where it falls and is cleared in the reset.
- **Coaching note:** cleanup motions are not part of the task. The spill is data, the cleanup is noise.

Violation: Bad pick, carry, or release

- **Visible cue:** an object is dragged along the table with no daylight under it, the gripper shifts or seats its hold again while the object is off the table, a part hangs tilted in the gripper because the grasp was taken off square, or an object is rolled past a quarter turn or turned over in the air.
- **SOP rule broken:** Steps 1 to 5, one object at a time, lifted straight up clear of the surface before any horizontal motion, gripped square on its sides, and tilted no more than a quarter turn.
- **Coaching note:** set it down to change the grip, and lift before the carry. A part that hangs tilted was taken off square.

Violation: Placement pushed, or correction limit exceeded

- **Visible cue:** a part, a bin, the pen, the rack or the sheet is shoved, nudged, or slid into position instead of being lifted clear and placed again, or a failed check is retried more often than its step allows.
- **SOP rule broken:** Steps 1 to 7, a bad placement is corrected by lifting the object clear and placing it again, and each check allows only the correction it states.
- **Coaching note:** nothing on this station is pushed into line. If grip or approach is forcing repeated correction, let review flag it and move on.

Violation: Dropped, collided, or knocked over

- **Visible cue:** under command, the gripper or a carried object strikes the scale, the pan, the rack, a bin or the sheet and knocks, tips, slides, or displaces it; or an object falls short of its spot, with no gripper failure, drift, or controller fault visible.
- **SOP rule broken:** Steps 1 to 7, keep each object on a clear travel path and release only after a settled placement.
- **Coaching note:** watch the whole path, not just the target. The scale sits in the middle of everything, so the route matters more than the reach.

Violation: Manipulation outside the camera frame

- **Visible cue:** a part pick, a set down on the pan, a lift off the pan, a reading or a mark happens partly or wholly outside the environment camera frame, or the arm blocks the camera's view of the display or of the sheet while the number is being read or written.
- **SOP rule broken:** Steps 1 to 7, every manipulation happens inside the camera frame and the display and the sheet stay visible.
- **Coaching note:** the work has drifted outside the framed area. Set the spots back to their setup positions and keep the arm clear of the display and the sheet.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a part still in the rack or still on the pan, a row unwritten or an extra mark on the sheet, a part in the wrong bin, the display showing anything but zero, the pen out of its holder, the arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 8, confirm the rack, the scale, the bins, the sheet and the tools, then return the arm home with its gripper open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arm does before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- Recording stopped or paused during the episode (recording system).
- Camera dropped frames or lost its feed (capture system).
- Hardware fault on an arm: gripper failure, drift, controller caused collision, or motor error.
- A spot sits outside the arm's comfortable reach, so the arm stalls at a joint limit or has to fold in on itself to work it. Move the spot and run the Setup checklists again.
- Defective scale: one that will not hold zero on an empty pan, drifts while a part sits still on it, reads a known weight wrong, or shows a number that will not settle.
- Scale display that is dim, flickering, cut off or unreadable on camera, so the number written down cannot be checked against it.
- A ZERO bar that does not respond to a single clean press, or that needs to be held down.
- Defective pan: one that is bent, not level, or fouled with something that will not wipe off.
- A table that rocks, or a draught, fan or vibration that moves the display off a settled number.
- Defective rack: one that slides, turns or lifts under a correct pick, or a slot that grips a part so a straight lift will not free it.
- Defective bin: one that will not stand upright in its holder, or that is dented or cracked.

- Dry pen, or a log sheet that will not wipe clean so a row or a box starts the episode already marked.
- A log sheet printed wrong: a missing band label, band boxes that touch, rows numbered other than 1 to 10, a weight box too small to hold a full number, or a band strip that does not match the bins.
- A part whose weight sits on the edge of a band, so no one can call the band from the number. Replace the part before the next episode.
- Grit, swarf or moisture picked up by a part during the episode that was not on it at the start and changes what it weighs.
- Rack holding fewer or more than ten parts at episode start, or the wrong mix of them. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Press the ZERO bar
2. Take one part from its rack slot
3. Set one part on the scale pan
4. Read the weight off the display
5. Take the pen from its holder
6. Write one weight on the log sheet
7. Mark one band box on the log sheet
8. Return the pen to its holder
9. Lift one part off the scale pan
10. Drop one part in the TOO LIGHT bin
11. Drop one part in the GOOD bin
12. Drop one part in the TOO HEAVY bin
13. Return the arm home and end the episode